Artificial Intelligence Lab (AL2002)

Date: 30/03/2026

Course Instructor(s)

Ms. Ramsha Jat

Lab Mid Exam (B)

Total Time: 90 minutes
Total Marks: 20
Total Questions: 03

Semester: spring-2026
Campus: Karachi
Dept: CS

Submission Instructions:

- You must comment your student ID on top of each file. (Line#1 of your code).
- Name the file for each question according to Roll_No e.g. **k24-xxxx_Q1.c, k24- xxxx _Q2.c** etc.
- Submission is via a client software so open the application present on the Desktop.
- Enter your username as **24K-xxxx** and its assigned password (**Default is Fast1234**).
- Submission is timed so after the time no submissi[illegible] ie accepted

CLO # 1: under[illegible]ing to real life applications of Agents and Environment

Q1. [7 marks] A smart home cleaning robot is designed to keep a room clean by moving across all areas and collecting dirt. The room is represented as a 4×4 grid, where each cell has a cleanliness score: **0** = dirty 1 = clean 2 = very dirty

The robot can move up, down, left, or right to an adjacent cell in one move. At each cell, it can "clean" the cell to collect the cleanliness score. The robot has a limited energy, which allows it to make a maximum of 8 moves.

The robot's goal is not just to reach a particular cell, but to maximize the total cleanliness score collected in the given number of moves. It must plan its path carefully to prioritize high-value cells while avoiding obstacles such as furniture or walls (marked as **X** in the grid).

Identify this agent as a Utility-Based Agent, since it evaluates possible outcomes and chooses actions that maximize total utility instead of only achieving a single goal.

S	2	0	0	1
0	X	1	2	0
0	2	0	X	0
0	0	1	0	2

CLO # 2: Explain and implement informed or uninformed search techniques such as Beam Search, to find optimal or near-optimal solutions in large search spaces.

Q2. [7 marks] A project manager is scheduling a set of tasks represented as nodes in a dependency graph. Some tasks depend on others and can only be completed after their predecessors. Each task has a priority value estimating its importance in achieving the project goal (lower value = higher priority).

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The manager uses Best-First Search to decide the order of executing tasks. At each step, the manager chooses the task with the highest priority (lowest heuristic value) among available tasks (whose dependencies are satisfied).

Your task is to simulate Best-First Search on the task graph and determine the sequence of tasks selected.

heuristics = {"T0": 5, "T1": 3, "T2": 4, "T3": 2, "T4": 1, "T5": 2,"T6": 0}

graph = {"T0": {"T1": 1, "T2": 1}, "T1": {"T3": 1, "T4": 1}, "T2": {"T5": 1}, "T3": {"T6": 1}, "T4":{"T6": 1}, "T5": {"T6": 1}, "T6": {}}

CLO # 3: Evolutionary local search techniques such as Beam Search, and Genetic Algorithms to find optimal or near-optimal solutions in large search spaces.

Q3. [6 marks] A software company is developing an optimization module that can automatically find the best input for a given mathematical function. You are tasked with finding the maximum value of the function $f(x)=x^2-3x+4$, where x is an integer between 0 and 15. To solve this, you decide to implement a Genetic Algorithm (GA).

The algorithm will represent potential values of x as binary chromosomes of 4 bits, and it will maintain a population of 4 individuals. The GA will use roulette wheel selection for parent selection, single-point crossover to produce offspring, and a mutation probability of 0.1 applied to each bit. Your goal is to simulate the GA to evolve a population that eventually contains the chromosome representing the maximum value of the function.

a. Encode a random initial population of 4 chromosomes in binary form.
b. Decode each chromosome into its decimal value and compute its fitness using the function f(x).
c. Perform one round of selection, crossover, and mutation, showing all intermediate steps, including selected parents, crossover points, and any mutations.
d. Identify the best chromosome after this generation and provide its decimal value and fitness.